Amir Reza **Isazadeh**

PhD in Rehabilitation Science

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Synopsis

Researcher and R&D Engineer specializing in **Tissue Biomechanics and Simulations**, with a track record of turning complex research models into **Clinically Viable Software and Medical Devices**.

Professional Profile

Researcher at Head and Neck Simulation Lab *Alberta, Canada*

University of Alberta ***2023–2025***

- **100× Performance Boost**: Developed a **real-time solver** that outperforms FEM-based optimization by two orders of magnitude and ensures clinically viable surgical designs (Open-source).
- **First-of-its-kind Soft Tissue VSP Pipeline**: Engineered a robust, **fully automated, end-to-end simulation pipeline** for large-scale surgical outcome prediction and individualized planning (Open-source).
- **Millimeter-Level Accuracy**: Designed an additively manufactured **surgical guiding toolkit** that delivers high precision when translating VSP into surgical execution.
- **≈95% Reduction in Preprocessing Time**: Slashed MRI-to-simulation workflow from **4** hours to **15** minutes with **AI-powered image segmentation**, enabling same-day individualized surgical planning.
- **Accelerated Clinical Adoption**: Partnered with surgeons and experts at UofA and Misericordia Hospitals to translate 40+ hours of OR shadowing into iterative, clinically driven VSP improvements.
- Secured a total of **$270K** in competitive scholarships and research funding through strategic proposals and earned a nomination for the prestigious **Vanier Scholarship**.
- Completed four impactful studies ahead of schedule, yielding a **PhD in under three years**.
- Served as a **reviewer for *IEEE*** and **mentored junior researchers** across diverse projects.

Research Assistant *Tehran, Iran*

Center of Excellence in Design, Robotics, and Automation ***2021***

- Developed a **rapid optimization framework** for automotive manufacturing process design; published the methodology in high-impact journals.
- **Led a team of four** to pilot the workflow with IKCO (automotive manufacturing); integration assessment projected $\approx$ 35% reduction in defective parts by improving upfront process design decisions.

Educational Excellence

PhD in Rehabilitation Science *Alberta, Canada*

University of Alberta ***2023–2025***

Thesis: Patient-Specific Virtual Surgical Planning for Tongue Reconstruction: Biomechanical Optimization, Surgical Simulation, and Guided Execution **GPA: 3.95**/4.0

M.Sc. in Mechanical Engineering *Tehran, Iran*

Sharif University of Technology ***2019–2021***

Thesis: Development of a Rapid Optimization Method and Its Application in Sheet Metal Forming Process Design **GPA: 4.0**/4.0

B.Sc. in Mechanical Engineering *Tehran, Iran*

Amirkabir University of Technology ***2013–2018***

Thesis: Algorithmic Design of Axial-Flow Turbomachinery Compressors **GPA: 3.9**/4.0

Publications

1. **Isazadeh A**, Chen S, Seikaly H, Zheng B, Westover L, Aalto D (**2025**) Guided tongue reconstruction: Feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model. *Surgical Oncology*. (under review).
2. **Isazadeh A**, Westover L, Seikaly H, Aalto D (**2025**) Computational analysis of tongue reconstruction surgery: The impact of free flap site and volume on the tongue anatomy and biomechanics. *PLOS ONE*. (under review).
3. **Isazadeh A**, Zenke J, Westover L, Seikaly H, Aalto D (**2025**) Patient-specific virtual surgical planning for tongue reconstruction: Evaluating hyperelastic inverse FEM with four simulated tongue cancer cases. *Phys Med Biol* 70:145017. doi: 10.1088/1361-6560/adebd8
4. **Isazadeh A** and Aalto D (**2025**) VSP-BioSim: Patient-Specific Virtual Surgical Planning via Biomechanical Simulations (Open-source software). doi: 10.5281/zenodo.17438091
5. **Isazadeh A** and Aalto D (**2025**) hiFEM: Hyperelastic Inverse Finite Element Method (Open-source software). doi: 10.5281/zenodo.17427896
6. **Isazadeh A**, Seikaly H, Westover L, Aalto D (**2024**) Algorithmically designed flaps in tongue reconstruction: A feasibility analysis. *Int J Comput Assist Radiol Surg* 19, 735–746. doi: 10.1007/s11548-024-03062-w
7. Shamloofard M, **Isazadeh A**, Shirin MB, Assempour A (**2022**) Optimum design of middle-stage tool geometry and addendum surfaces in sheet metal stamping processes using a new isogeometric-based framework. *Proc Inst Mech Eng Part B J Eng Manuf* 236:741–757. doi: 10.1177/09544054211041046
8. **Isazadeh A**, Shamloofard M, Assempour A (**2021**) Some improvements on the one-step inverse isogeometric analysis by proposing a multi-step inverse isogeometric methodology in sheet metal stamping processes. *Applied Mathematical Modeling* 91:476–492. doi: 10.1016/j.apm.2020.09.032

Conference Presentations

- Virtual planning of tongue cancer surgery: patient-specific simulation of hemiglossectomy biomechanics. *19th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE)*. Vancouver, Canada. 2024.
- Algorithmically designed flaps in tongue reconstruction: A feasibility analysis. *Prairie Collaborative Research Conference*. Edmonton, Canada. 2023.
- Analysis of a sheet metal stamping process with high drawing depth using a new NURBS-based methodology. *28th International Conference of Mechanical Engineers Association of Iran*. Tehran, Iran. 2020.

Recognition, Awards & Scholarships

- Graduate Student Support Award ($5,000) *Canada*
  GPS Awards, University of Alberta *2024*
- Nominated for Vanier Canada Graduate Scholarships *Canada*
  CIHR Federal Competition *2023*
- Mickleborough Scholarship in Rehabilitation Science ($1,000) *Canada*
  University of Alberta *2023*
- Faculty of Rehabilitation Medicine GRAF ($64,000) *Canada*
  University of Alberta *2023*
- Graduate Recruitment Scholarship ($5,000) *Canada*
  University of Alberta *2023*
- Industrial Transformation and Training Centre for Joint Biomechanics Scholarship ($99,000) *Australia*
  Queensland University of Technology (successful, declined to pursue PhD at UofA) *2022*
- Higher Degree Research Tuition Fee Sponsorship ($96,000) *Australia*
  Queensland University of Technology (successful, declined to pursue PhD at UofA) *2022*
- Recognized as "Outstanding Student" and ranked in the top 5% of graduates, Mechanical Engineering Department *Iran*
  Sharif University of Technology *2021*
- Ranked 11th (out of 13,900) in the Iranian national M.Sc. entrance exam *Iran*
  National Organization of Educational Testing *2018*
- Qualified for double B.Sc. degree program, Exceptional Talents Office *Iran*
  Amirkabir University of Technology *2015*
- Tuition waiver scholarship for B.Sc. program *Iran*
  National Organization of Educational Testing *2015*
- Ranked in the top 1% in the Iranian national B.Sc. entrance exam *Iran*
  National Organization of Educational Testing *2013*

Technical Mastery

Computational Mechanics & Simulation

- Nonlinear Finite Element Method (FEM), inverse FEM
- Soft Tissue Biomechanics, Large Deformations, Visco-hyperelastic Modeling
- Real-time Simulations; NVIDIA Warp, Omniverse, Isaac Sim

Virtual Surgical Planning

- AI-Driven MRI Segmentation
- Automated Patient-Specific Model Preprocessing Pipeline

Tools

- $FEM\ solvers$: FEBio, ABAQUS, PolyFEM, NVIDIA Warp
- $Mesh\ generation$: fTetWild, Interactive All-hex Meshing, HexBox, and PolyCube libraries
- $Geometric\ processing$: Open3D, PyVista
- $Geometric\ modeling$: SolidWorks, MeshMixer, Rhino/Grasshopper, Autodesk Fusion360, Autodesk Inventor
- $Programming$: Python, MATLAB
- AI/ML : Scikit-learn, PyTorch
- $Optimization$: Genetic Algorithms, Neural Networks
- $Medical\ image\ segmentation$: Slicer 3D, Biomedisa (AI-powered)
- $Statistics$: statsmodels, R, Seaborn, STATA
- $Data\ visualization$: Seaborn, Matplotlib, Paraview
- $Version\ control$: Git, GitHub
- Attended numerous workshops such as Biomedisa, ArtiSynth, FEBio, etc.

Teaching & Outreach

- Volunteered for MRI pilot studies — University of Alberta 2023
- Teaching Assistant — Metal Forming Processes and Strength of Materials 2019–2020
- Co-founded ”Konkursite”, a non-profit organization that provided online education to over 8,000 underserved students 2017–2020